ONTI · ONTOLOGY ANALYSIS REPORT

Full Analytics Report | Scenario: VNB / APE Insurance Sample Pack | Data: Seed: Insurance Products, Seed: Channel-Product Mix, Seed: Distribution Channels, Seed: Agents (+7 more)

Generated: 2026-10-13T06:54:49+00:00
Ontology pack: VNB / APE Insurance Sample Pack

– Business Year	– Stated Total APE	– Stated Total VNB
– Stated VNB Margin	**$7.42M** Derived Total APE	**$1.45M** Derived Total VNB

Date: 2026-10-13 | **Ontology pack:** VNB / APE Insurance Sample Pack | **Scope:** Pack | **Questions analysed:** 8

1. Executive Summary

- Ontology context — Ontology narrative _Narrative template error: a number is required_
- Analysed **8 question(s)** (8 rule-engine, 0 AI-assisted).
- Key figures: **Business Year –**, **Stated Total APE –**, **Stated Total VNB –**, **Stated VNB Margin –**.
- Highest Unit-Linked at 25.0%; lowest Term Life at 10.0%. Grouped bars compare VNB Margin, Commission Rate, Expense Ratio.
- Highest item 2 at 66.0%; lowest item 4 at 5.0%. Grouped bars compare Mix Weight, Year.

2. Q&A Record

1. What is the Business Year?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

2. How is Business Year calculated?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

3. What is the Stated Total APE?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

4. How is Stated Total APE calculated?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

5. What is the Stated Total VNB?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

6. How is Stated Total VNB calculated?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

7. What is the Stated VNB Margin?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

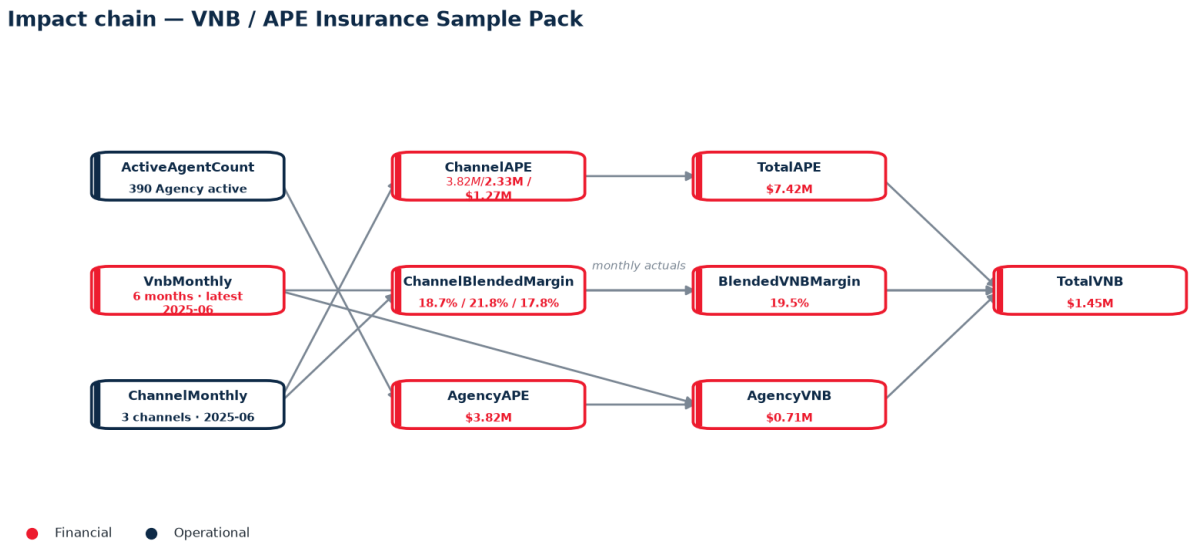
8. How is Stated VNB Margin calculated?

*Mode: **Rules Engine***

No rule matched — ask this in AI mode for a model answer grounded in the same knowledge base.

3. Ontology Evidence — Impact Chain

The diagram below shows the directed reasoning chain linking operational drivers to KPI outcomes captured by the analysed questions.



Impact chain — VNB / APE Insurance Sample Pack

Entity & attribute definitions

Entity	Layer	Attributes
Agent	3	Agent Channel ID, Agent ID, Agent Productivity, Certification, Churn Risk, Tenure (Months)
VNB Monthly Results	1	Month, Total APE, Total VNB, VNB Margin, Agency APE, Agency VNB, Agency share of VNB, MoM VNB change
Channel Product Mix	3	Channel-Product APE, Channel-Product VNB, Mix Channel ID, Mix ID, Mix Product ID, Year, Product Mix %
Channel Monthly KPIs	2	Month, Channel ID, Active agents, Avg productivity, APE, Weighted VNB margin, VNB
Distribution Channel	3	Agent Churn Rate, Agent Count, Average Productivity, Channel ID, Channel Name, Channel Type
Agent Monthly Performance	3	Month, Agent ID, Channel ID, Status, Productivity, APE, Churn risk
Insurance Product	3	Commission Rate, Expense Ratio, Product ID, Product Name, Product Type, Risk Discount Rate, VNB Margin

Entity	Layer	Attributes
Product Mix Monthly	3	Month, Channel ID, Product ID, Product mix, VNB margin, Weighted margin component
VNB Summary	3	Overall VNB Margin, Total APE, Total VNB, Year
Scenario Analysis	4	Scenario / input, Value, Unit / interpretation

Metric formulas (calculation basis)

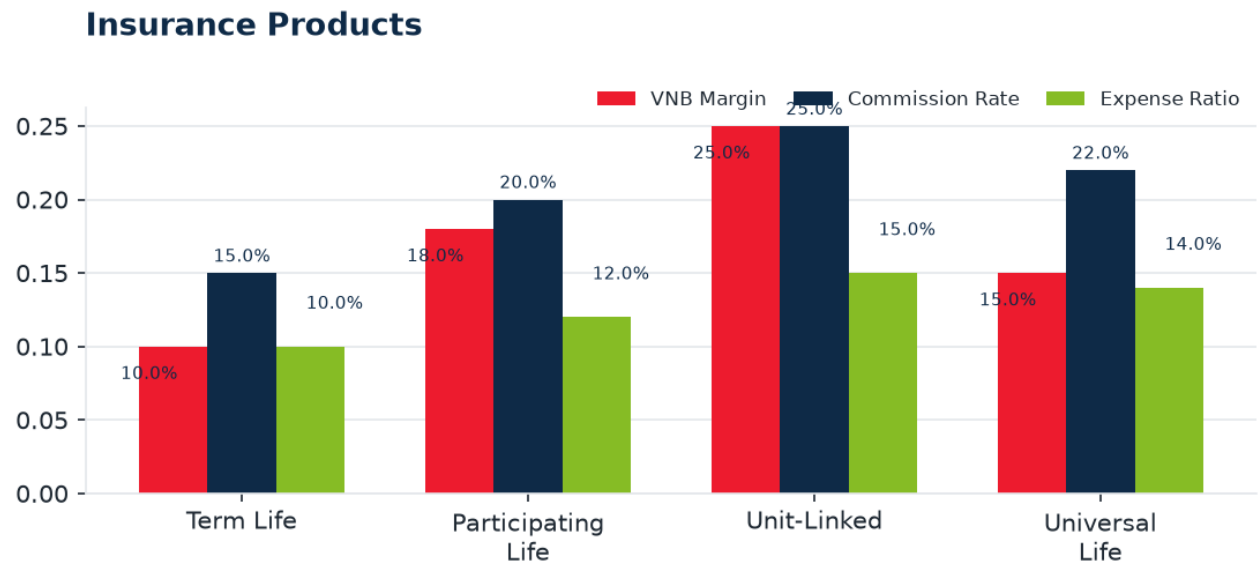
Metric	Formula
Business Year	`cell('kpi', 'year')`
Stated Total APE	`cell('kpi', 'ape')`
Stated Total VNB	`cell('kpi', 'vnb')`
Stated VNB Margin	`cell('kpi', 'margin')`
Derived Total APE	`round(sum(rows('channels'), 'ape'), 2)`
Derived Total VNB	`round(sum(rows('channels'), 'vnb'), 3)`
Derived VNB Margin	`round(derived_vnb / derived_ape, 4)`
VNB Gap	`round(derived_vnb - stated_vnb, 2)`
Margin Gap	`round(derived_margin - stated_margin, 4)`

4. KPI and Numbers

Metric	Value	Unit
Business Year	–	
Stated Total APE	–	$M
Stated Total VNB	–	$M
Stated VNB Margin	–	%
Derived Total APE	**$7.42M**	$M
Derived Total VNB	**$1.45M**	$M
Derived VNB Margin	**19.5%**	%
VNB Gap	–	$M
Margin Gap	–	%

5. Chart Analysis

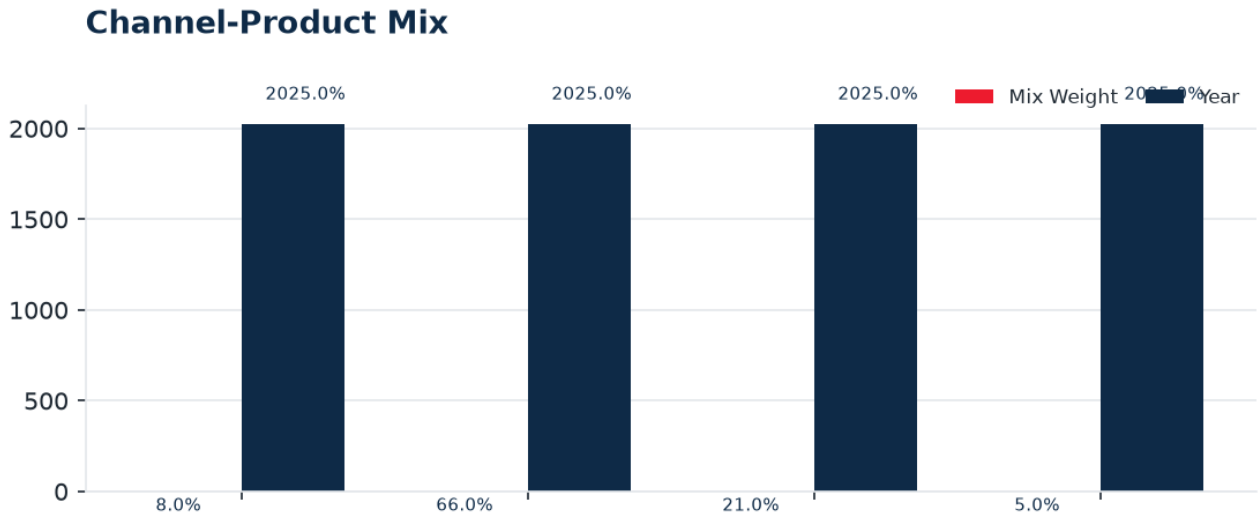
1. Insurance Products



Insurance Products

Highest **Unit-Linked** at 25.0%; lowest **Term Life** at 10.0%. Grouped bars compare VNB Margin, Commission Rate, Expense Ratio.

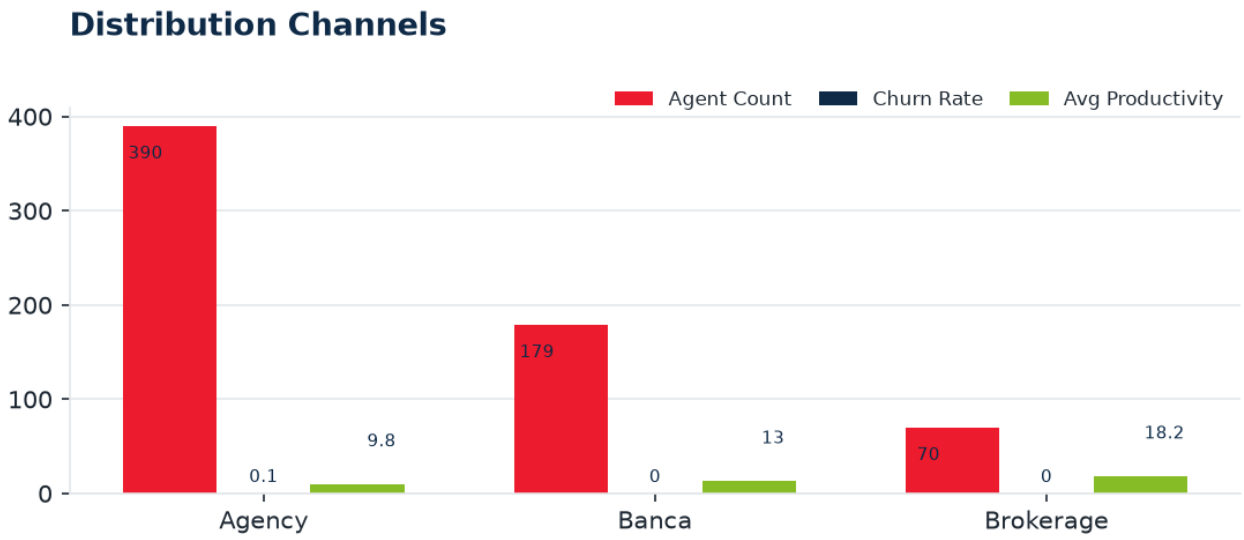
2. Channel-Product Mix



Channel-Product Mix

Highest **item 2** at 66.0%; lowest **item 4** at 5.0%. Grouped bars compare Mix Weight, Year.

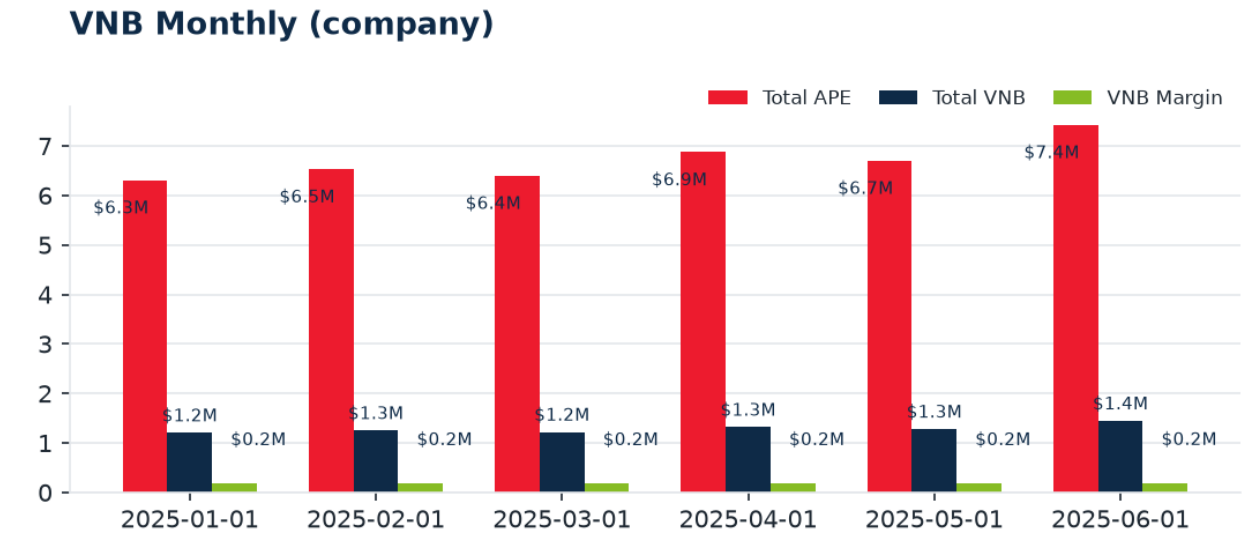
3. Distribution Channels



Distribution Channels

Highest **Agency** at 390; lowest **Brokerage** at 70. Grouped bars compare Agent Count, Churn Rate, Avg Productivity.

4. VNB Monthly (company)

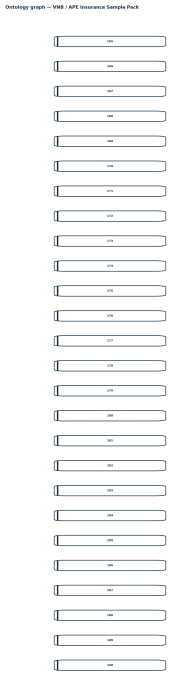


VNB Monthly (company)

Highest **2025-06-01** at $7.4M; lowest **2025-01-01** at $6.3M. Grouped bars compare Total APE, Total VNB, VNB Margin.

6. Ontology Knowledge Graph

The full entity-relationship graph of the active ontology pack — the same structure that powers Q&A, the dashboard and the analytics above.



Ontology graph — VNB / APE Insurance Sample Pack

7. Appendix

Methodology & assumptions

- Answers are produced by Onti's rule engine against the active ontology pack; AI answers are grounded with the same knowledge base and must not introduce external facts.
- Derived columns and metric formulas are evaluated in declaration order from the active dataset; the calculation basis is listed in the Ontology Evidence section.
- Charts render only tables with 2-15 rows in their active dataset; sections are omitted when their source data is absent.

Dataset inventory

Table	Entity	Active dataset	Rows
Company KPI (year)	vnb_summary	–	0
Insurance Products	insurance_product	Seed: Insurance Products	4
Channel-Product Mix	channel_product_mix	Seed: Channel-Product Mix	4
Distribution Channels	distribution_channel	Seed: Distribution Channels	3
Agents	agent	Seed: Agents	500
VNB Monthly (company)	vnb_monthly	Seed: VNB Monthly (company)	6
Channel Monthly (KPIs)	channel_monthly	Seed: Channel Monthly (KPIs)	18
Product Mix Monthly	product_mix_monthly	Seed: Product Mix Monthly	24
Agent Master	agent	Seed: Agent Master	500

Table	Entity	Active dataset	Rows
Agent Monthly Performance	agent_monthly	Seed: Agent Monthly Performance	3000
KPI Linkage (ontology)		Seed: KPI Linkage (ontology)	7
Scenario Analysis (seed)	scenario_analysis	Seed: Scenario Analysis (seed)	15

Generation metadata

Item	Value
Report ID	`5b2428c75d19`
Generated at (UTC)	2026-10-13T06:54:49+00:00
Sections included	summary, qa, evidence, kpi, charts, ontology, appendix
AI interpretation	off